

# **Build a Large Language Model from Scratch**

*this thesis is a summary of [1] and reports on the experience with it.*

## **REPORT 1**

Understanding large language models

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### **Structure**

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## Introduction

Nowadays, technology and innovations are advancing rapidly, leading to useful and very interesting new creations. One of the most remarkable is the LLM.

**Large Language Model** is a deep neural network specifically created to understand, generate, and interact with human language. It's trained on a significant portion of the text available on the internet.

The "large" in its name is critical. It refers to two things:

1. **The immense scale of the training data.**
2. **The millions, or even billions, of internal parameters** it uses to make predictions.

To better understand what the LLM is, first we need to understand how it's connected to or differentiated from broader concepts, starting from AI

**AI** unites all the systems that can perform tasks requiring human-like intelligence. **Machine Learning** is the Toolkit. It's a wide field of algorithms that can learn from structured data to perform a specific task, like prediction or classification. **Deep Learning** is the Engine. It's a specialized part of that toolkit that uses deep neural networks to automatically discover relevant patterns from raw, unstructured data. And finally, a **Large Language Model (LLM)** is the Application. It is a product built using the Deep Learning engine. It's hyper-specialized for one thing: mastering language. While a general Deep Learning model might classify images, an LLM is dedicated to the sequential, contextual, and generative nature of text.

## Core Concepts

Following the foundational overview, In order to get comprehensive and capable LLM, there is two-stage process to go through. It's developed from a generalist into a specialist.

### Stage 1: Pretraining

The model is trained on a massive amount of unlabeled text (like books and websites) with a simple objective to predict the next word. That is exceptionally resource-intensive and expensive, but as a Result, we get a powerful but general-purpose Foundation Model.

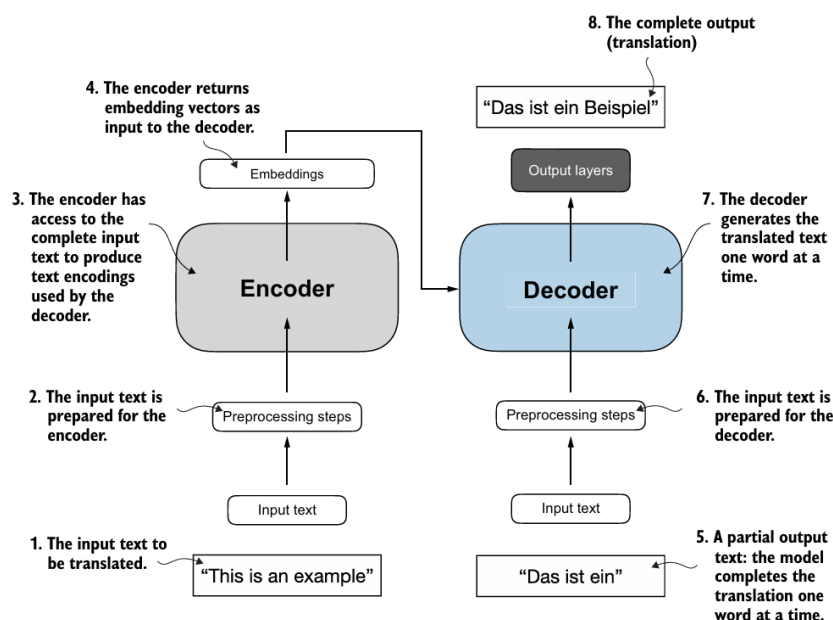
### Stage 2: Fine-Tuning

The model is further trained on a much smaller, labeled dataset tailored to the desired skill.

## Transformer Architecture

The original **Transformer** model's basic structure consists of two parts: an **encoder** and a **decoder**. As can be seen in the figure below, the encoder reads the input text and converts it into a vector representation. It encodes its contextual information. The decoder then decodes

this numerical representation to generate the output text. One of the major uses of this model is a translation application, whereby the encoder would process the source language text, and the decoder would generate the translated text in the target language.



**Fig.1: Transformer Architecture**  
(Source: [1], p. 8)

Moreover, the self-attention mechanism is one of the key features of the transformer architecture. The mechanism is employed to provide weights for different words so that the model can acquire information regarding dependencies and word relations.

The **GPT (Generative Pre-trained Transformer)** architecture simplifies this design. It utilizes **only the decoder** portion of the original Transformer. This focus on the decoder allows them to predict next word in a sequence, which is the core of text generation.

## Technical Insights

the most remarkable outcomes of LLM development is how a model trained on a simple task develops complex, un-taught skills, and why this makes it so versatile.

A key question is how a model like GPT, trained simply to predict the next word, can perform tasks like translation. which is **emergent behavior**, that happened due to large amount of data and ability of algorithm to find patterns and connections withing them. It essentially discovered these abilities on its own.

This emergent capability is best demonstrated by two concepts:

- **Zero-Shot Learning:** This is the ability to perform a task without seeing any prior examples. You can give the model a completely new instruction.

- **Few-Shot Learning:** This involves providing the model with a few examples of a task before requiring answer.

## Real-World Relevance

Large language models are here to stay as a boost modern productivity, having literally transformed the nature of work. The existence of coherent text generation, document summarization and sophisticated translation has increased global communication and information production. not just human language, LLMs speed up software creation by producing code as well as finding where bugs might be around. Moreover, in specialized, knowledge-driven sectors such as finance and medicine, they serve as formidable aides.

Finally, as already stated, the fact that LLMs are to be found everywhere today highlights their all-encompassing ability to organize and interpret data streams, which has granted LLMs a unique status in terms of optimizing efficiency and expertise in almost every field.

## Reflection

Based on this initial research, we gained a solid foundational understanding of Large Language Models and the two-stage process it needs: massive **pretraining** and **fine-tuning**. Most importantly, we understand that an LLM's most impressive abilities, like translation, aren't directly programmed. They are **emergent behaviors** that arise naturally from the model learning deep statistical patterns in language simply by being trained to predict the next word.

### Our Goals and What's Next

So far our understanding is theoretical. **From Now on our primary goal is to read and understand remaining parts from book “build a Large Language Model from scratch”**

## References

[1] Raschka, S. (2024). *Build a large language model (from scratch)*. Shelter Island, NY: Manning Publications.